Fairness-Aware Bandits for Network Routing in Quantum & Clinical Settings

via Multi-Armed, Contextual, and Informative Contextual Decision Spectra

Piter Z. Garcia
MS Data Science / Decision-Making & Algorithmic Fairness
Rochester Institute of Technology (RIT)
pizg8794@g.rit.edu

Daniel Krutz, Travis Desell
Department of Software Engineering, Data Science
Rochester Institute of Technology (RIT)
dxkvse@g.rit.edu, tjdvse@g.rit.edu

Abstract*—*[modified: Sequential decision systems increasingly determine how scarce diagnostic, computational, and communication resources are routed under uncertainty. This study examines fairness-aware bandits across a decision spectrum, from non-contextual multi-armed bandits to contextual and informative contextual bandits, for settings where context is missing, delayed, noisy, or group-dependent. The central claim is that quantum routing and clinical diagnostic routing share a common sequential resource-allocation structure: policies must choose actions from partial feedback, yet groups or flows with the least reliable context can absorb the worst errors.]

[newtext: The work develops a reusable evaluation framework for this bandit spectrum across two executable environment families. The quantum domain builds on a state-preserving quantum-routing evaluation stack, while the clinical domain uses an embedded synthetic clinical environment with resource allocation, clinical models, and EQUITAS-mediated fairness reporting. Preliminary quantum results show that predictive-context policies improve over no-context baselines by 3.74 percentage points internally and 2.83 percentage points externally. In the clinical setting, the embedded EQUITAS-aware allocator improves oracle-normalized efficiency from 75.09% to 91.23%, raises mean success from 50.0% to 72.5%, and reduces the latency parity gap from 18.0 to 10.17 relative to the baseline. In the related bioinformatics proxy, context-aware scoring increases predicted-pair output from 14.96 $\pm$ 10.45 to 61.36 $\pm$ 44.21, with optimized output reaching 66.07 $\pm$ 46.36 and average disparate-impact improvement of +0.805 after augmentation. Together, these components define a method for measuring and mitigating time-evolving fairness disparities while preserving utility under non-stationarity and limited context.]

***Index Terms*—Fairness-aware bandits, contextual bandits, resource allocation, clinical routing, quantum routing, partial feedback, missing context, algorithmic fairness.**

I. Introduction

[modified: Many applied data science systems are sequential decision systems that route limited resources under uncertainty: which diagnostic test to order, which case to escalate, which model or pipeline to trust, or which quantum-network path should receive scarce entanglement-generation attempts. These decisions are difficult because the system observes only partial feedback. The chosen action reveals an outcome, but the unchosen alternatives remain counterfactual. They are also fairness critical because the quality of observed context is often uneven across clinical groups, sites, and cohorts or across quantum flows and service classes. When one population has delayed testing, incomplete patient history, lower-quality measurements, or less representative calibration data, a policy optimized only for aggregate reward can hide subgroup errors and compound disparities over time; see Figure 1. The same structure appears in quantum routing when flows depend on uneven link, fidelity, capacity, or route-state evidence; see Figure 6.]

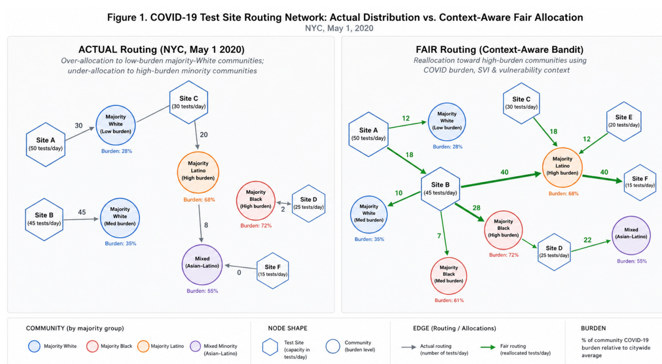

Fig. 1. Real-world clinical routing illustration. The left side shows weak-context allocation, while the right side shows context-aware fairness allocation under the same shared routing problem.

[modified: This work frames the scarce-resource routing problem as fairness-aware bandit learning across a decision spectrum. At each round, a policy observes available context, chooses an action, receives reward only for the chosen action, and updates its future decisions. The scientific questions focus on whether movement along the bandit spectrum reduces disparities when context is missing, delayed, noisy, or group-dependent. This matters in clinical workflows because false-negative, delayed-escalation, or under-testing gaps can produce direct harm. It also matters in quantum-network routing because future communication systems will allocate scarce probabilistic resources among competing flows; fairness in success probability and latency is a service-quality requirement.]

The study develops a shared environment–runner–evaluator architecture for two complementary routing domains. The first is a quantum-routing testbed based on hybrid contextual bandits for entanglement routing and qubit allocation [1].

It exposes noisy links, uncertain entanglement generation, adversarial or non-stationary conditions, allocator decisions, and saved evaluator state. The second is an embedded clinical simulation motivated by equitable bioinformatics and fairness auditing work [2]. It models test ordering, retesting, escalation, diagnostic attention, model-assisted triage, site-level constraints, and cohort-level context quality. The shared abstraction is that each domain routes scarce resources under partial feedback while fairness risk emerges from unequal context quality.

[newtext: Four scientific research questions drive this work. RQ1 asks how much quality context improves utility and fairness relative to non-contextual baselines. RQ2 asks how quickly disparities emerge when a group, site, cohort, or flow class receives noisier, delayed, or incomplete context. RQ3 asks whether fairness-mediated policy updates reduce outcome gaps without unacceptable utility loss. RQ4 asks whether the same fairness mediation strategy transfers between clinical diagnostic simulation and quantum routing, or whether mitigation must remain domain-specific. Our preliminary evidence supports the context-gain premise: predictive-context quantum policies improve over no-context baselines by 3.74 percentage points internally and 2.83 percentage points externally. In the embedded clinical setting, the EQUITAS-aware allocator improves oracle-normalized efficiency from 75.09% to 91.23%, raises mean success from 50.0% to 72.5%, and reduces the latency parity gap from 18.0 to 10.17 relative to the baseline.]

II. Related Work

A. Bandits and Contextual Sequential Decision-Making

Multi-armed bandits formalize sequential learning under partial feedback: a policy repeatedly chooses among actions, observes reward only for the selected action, and must balance exploration with exploitation. UCB-style algorithms provide finite-time regret guarantees in stochastic settings [3], while Thompson sampling offers a Bayesian alternative with strong empirical performance [4]. The broader bandit literature supplies the regret, uncertainty-estimation, and baseline policy machinery used in this study [5].

Contextual bandits extend this setting by conditioning action selection on observed side information. LinUCB and related linear contextual methods show how context can improve sequential decision quality [6], [7], and reduction-based approaches scale contextual bandits to richer policy classes [8]. However, much of this work optimizes expected reward under the assumption that context is observed accurately, promptly, and representatively. That assumption is fragile in resource-allocation settings where context can be missing, delayed, noisy, or systematically lower quality for particular groups or flows.

Offline policy evaluation is relevant because fairness-aware routing systems often combine logged simulator outputs with fresh online runs. Doubly robust estimators combine reward modeling with inverse-propensity weighting and can reduce variance when evaluating bandit policies from logged data [9]. These methods motivate careful logging of actions, selection rules, context snapshots, outcomes, and oracle comparisons. They also expose a fairness concern: if a subgroup has weaker support in the logged data, group-specific estimates can remain noisy even when aggregate estimates appear stable.

B. Fairness, Missing Context, and Non-Stationarity

Fairness in supervised learning is often expressed through group-conditioned error criteria such as equality of opportunity and equalized odds [10]. Sequential settings complicate these criteria because fairness constraints affect exploration, data collection, and the future evidence available to the learner. Joseph et al. connect fairness directly to bandit learning [11], while Chen et al. study fair contextual multi-armed bandits with explicit group constraints [12]. Offline contextual-bandit fairness work further shows how importance-weighted estimators can support high-probability constraints [13].

A separate line of work addresses non-stationarity by adapting to changing reward distributions through discounting, change detection, or variation budgets [14], [15]. Restricted- or unobserved-context bandit methods similarly study decision-making when features are unavailable, latent, or costly to observe [16], [17]. These literatures address important parts of the problem, but usually in isolation: standard bandits emphasize regret, fairness work emphasizes disparity constraints, non-stationary bandits emphasize reward drift, and missing-context work emphasizes feature availability. The present study treats missing context, time-varying reward, and group disparity as simultaneous properties of one sequential decision process.

C. Clinical Fairness and Quantum Routing

Clinical algorithmic bias studies show that proxy targets and measurement processes can encode structural inequities, producing under-allocation for some groups even when aggregate performance appears strong [18]. This motivates outcome-based monitoring of false-negative gaps, false-positive gaps, escalation delays, and other group-conditioned harms. In quantum-network routing, decisions are governed by probabilistic entanglement generation, fidelity constraints, storage limits, and scarce network resources [19]–[21]. Existing quantum-routing work primarily emphasizes throughput, fidelity, latency, or robustness. Service equity across flows or user classes remains less developed.

The clinical and quantum settings therefore expose a shared research gap. Both involve sequential allocation under uncertainty, partial feedback, and uneven context quality, but existing methods rarely evaluate outcome-based fairness under missing context and distribution shift across both domains. This gap motivates the shared missing-context routing formulation used in this study.

Table I summarizes how these bodies of work define the technical foundation and remaining gap. Existing methods provide regret guarantees, contextual learning rules, fairness definitions, non-stationary adaptation, and domain-specific routing models, but they rarely connect these pieces into one evaluation framework for fairness-aware routing under missing context. That connection is the role of the proposed clinical–quantum formulation.

TABLE I
LITERATURE AREAS SUPPORTING THE SHARED MISSING-CONTEXT ROUTING FORMULATION.

Literature area	Role in this study
Bandit foundations	partial feedback, regret, and baseline policies
Contextual bandits	side information and missing-context sensitivity
Fairness definitions	outcome gaps, FNR/FPR, and service equity
Non-stationarity	reward drift and time-varying disparity analysis
Clinical bias studies	subgroup error monitoring and proxy-risk caution
Quantum routing	resource-scarce routing testbed and simulator grounding

III. Approach and Methodology

A. Shared Missing-Context Routing Model

Both domains are modeled as routing under missing context. The problem formulation identifies the shared harm mechanism: clinical and quantum systems route scarce resources under uncertainty, and unequal context quality can produce repeated allocation harm. Data characterization identifies the measurable patterns that make that harm testable, including group, site, cohort, service line, test distribution, escalation action, latency, reward, success, scenario, allocator, route choice, and evaluator-state fields.

In the clinical domain, the routed resource is diagnostic attention: scarce tests, retests, triage escalation, model-assisted prioritization, or model/pipeline selection. In the quantum domain, the routed resource is entanglement-generation effort, qubit allocation, or path capacity. In both domains, the policy observes partial context, chooses an action, receives partial feedback, and updates. The key experimental variable is the richness and reliability of context available to the policy.

Figure 2 summarizes the shared model. Phase 1 defines the environment and available resources. Phase 2 determines what context is observed and which context is missing or degraded. Phase 3 selects an action using a non-contextual, contextual, or informative contextual bandit policy. Phase 4 evaluates both performance and fairness, then feeds mitigation or updated policy information back into future decisions. This model is useful because it makes the failure point explicit: if context is systematically lower-quality for one group or flow class, the learning process can convert measurement inequity into repeated allocation inequity. It also clarifies where the experiments intervene: missingness is injected before the policy sees the state, while mitigation is evaluated only after utility and disparity have been logged.

At round t, the learner receives an observed context vector $\tilde{\mathbf{x}}_t$, which may differ from the latent complete context $\mathbf{x}_t$ because of missingness, delay, or measurement noise. It chooses an action a_t and observes reward $r_t(a_t)$ only for that action. Utility is tracked through cumulative regret,

$$R_T = \sum_{t=1}^{T} \left(r_t(a_t^*) - r_t(a_t) \right), \tag{1}$$

where a_t^* is the best available action under the simulator oracle. Fairness is tracked through time-indexed disparity gaps,

$$\Delta_m(t) = \left| m_t(G=0) - m_t(G=1) \right|, \tag{2}$$

where m_t is a domain-specific outcome metric such as false-negative rate, success probability, latency, or escalation delay. This notation separates the learning objective from the fairness audit: a policy can have low regret in aggregate while still producing high $\Delta_m(t)$ for specific groups.

Algorithm 1 Context-Spectrum Bandit Runner

Input: Environment $\mathcal{E}$; action set $\mathcal{A}$; policy $\pi \in \{\pi_0, \pi_1, \pi_2\}$; horizon T.
Output: State log $\mathcal{L}$ and policy-specific utility summaries.
Initialize policy state $\mathcal{B}_\pi$ and empty log $\mathcal{L}$.
Bandit constraint: only the selected action a_t reveals $r_t(a_t)$; unselected actions remain counterfactual.
for $t = 1, \ldots, T$ **do**
Observe available actions $\mathcal{A}_t$, observed context $\tilde{\mathbf{x}}_t$, optional predictive context $\hat{\mathbf{x}}_t$, and context-quality indicators q_t for logging and optional context recovery.
Build policy input

$$z_t = \begin{cases} \emptyset, & \pi = \pi_0 \quad \text{(MAB)}, \\ \tilde{\mathbf{x}}_t, & \pi = \pi_1 \quad \text{(CMAB)}, \\ [\tilde{\mathbf{x}}_t, \hat{\mathbf{x}}_t], & \pi = \pi_2 \quad \text{(iCMAB)}. \end{cases}$$

for each $a \in \mathcal{A}_t$ **do**
Score action a using $S_t(a \mid z_t; \mathcal{B}_\pi)$.
end for
Select

$$a_t \leftarrow \arg\max_{a \in \mathcal{A}_t} S_t(a \mid z_t; \mathcal{B}_\pi).$$

Execute a_t and observe reward $r_t(a_t)$, outcome o_t, and group/flow label.
Update $\mathcal{B}_\pi$ using only selected-action feedback $(z_t, a_t, r_t(a_t), o_t)$.
Log $\ell_t = \langle z_t, q_t, a_t, r_t(a_t), o_t, \text{group/flow} \rangle$; append ℓ_t to $\mathcal{L}$.
end for
return $\mathcal{L}$ and policy-specific utility summaries.

B. Algorithmic Context Spectrum

[newtext: The implementation uses the same runner, action set, reward logging, and fairness audit across the context spectrum. What changes is the evidence passed into the action scorer. MAB uses action history only, CMAB adds observed context, and iCMAB augments observed context with predictive or recovered evidence. In this sense, iCMAB acts as a context-side mediator before scoring, whereas EQUITAS mediates fairness after utility scoring. Algorithm 1 shows this shared runner, and Appendix E provides the side-by-side code-family trace.]

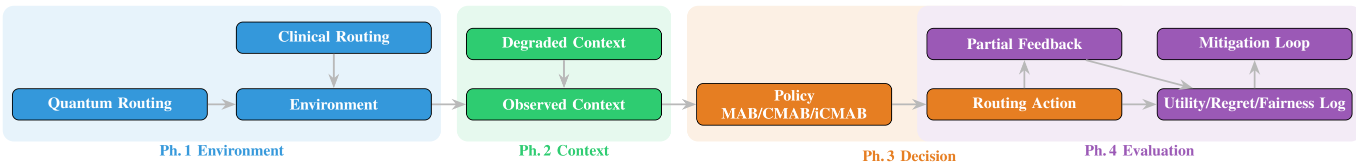

Fig. 2. Shared missing-context routing pipeline. The environment (clinical test ordering or quantum path selection) feeds an observation stage where context may be missing, noisy, or delayed; the policy family (MAB/CMAB/iCMAB) acts on the degraded signal; partial feedback is logged for utility, regret, and fairness; and a mitigation loop closes the cycle.

[newtext: Algorithm 1 links the method to the context-spectrum findings in Table VIII and Fig. 4. The runner, action set, logging, and fairness audit remain fixed; the ablation changes only whether the scorer receives no context, observed context, or predictive context.]

C. Bandit Policies and EQUITAS Mediation

The baseline non-contextual policy is an upper-confidence-bound bandit. For action a, let $N_{a,t}$ be the number of times action a has been selected before round t and let $\hat{\mu}_{a,t}$ be its empirical reward mean. The UCB score is

$$U_t^{\mathrm{MAB}}(a) = \hat{\mu}_{a,t-1} + \alpha\sqrt{\frac{2\log t}{\max(1, N_{a,t-1})}}, \tag{3}$$

and the learner selects $a_t = \arg\max_a U_t^{\mathrm{MAB}}(a)$. This baseline deliberately ignores patient, site, topology, load, or fidelity features, so it measures the cost of acting without context.

The contextual policy replaces scalar action means with action-specific linear models. For each action, let $A_{a,t}$ and $b_{a,t}$ denote the accumulated feature covariance and reward-weighted feature vector, with $\hat{\theta}_{a,t} = A_{a,t}^{-1} b_{a,t}$. Given observed context $\tilde{\mathbf{x}}_t$, the contextual UCB score is

$$U_t^{\mathrm{CMAB}}(a) = \tilde{\mathbf{x}}_t^\top \hat{\theta}_{a,t-1} + \alpha\sqrt{\tilde{\mathbf{x}}_t^\top A_{a,t-1}^{-1} \tilde{\mathbf{x}}_t}. \tag{4}$$

After observing reward $r_t(a_t)$, the selected action updates as $A_{a_t,t} = A_{a_t,t-1} + \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t^\top$ and $b_{a_t,t} = b_{a_t,t-1} + r_t(a_t)\tilde{\mathbf{x}}_t$.

EQUITAS adds fairness mediation as a governed scoring layer around either bandit family. Let $U_t(a)$ be the utility score from Eq. 3 or Eq. 4, $\widehat{\Delta}_{m,t}(a)$ be the estimated disparity gap that would result from action a, ϵ be an allowable gap, and $C_t(a)$ be a context-deficit penalty for missing, stale, or low-confidence features. The mediated score is

$$S_t^{\mathrm{EQUITAS}}(a) = U_t(a) - \lambda \max\left(0, \widehat{\Delta}_{m,t}(a) - \epsilon\right) - \gamma C_t(a), \tag{5}$$

with action $a_t = \arg\max_a S_t^{\mathrm{EQUITAS}}(a)$. The parameters λ and γ control the utility–fairness and utility–context-quality tradeoffs. In clinical routing, $\widehat{\Delta}$ tracks error, escalation, or delay gaps across sites and cohorts. In quantum routing, it tracks service gaps such as success probability, latency, or resource access across flows or service classes.

Algorithm 2 EQUITAS-Mediated Fairness-Aware Routing

Input: Environment $\mathcal{E}$; actions $\mathcal{A}$; policy $\pi \in \{\pi_0, \pi_1, \pi_2\}$; fairness metric m; gap ϵ; weights λ, γ; horizon T.
Output: State log $\mathcal{L}$, utility summaries, fairness summaries, and reporting datasets.
Initialize policy state $\mathcal{B}_\pi$, fairness accumulators $\mathcal{F}$, context-quality summaries $\mathcal{Q}$, and log $\mathcal{L}$.
for $t = 1, \ldots, T$ **do**
Observe $\tilde{\mathbf{x}}_t$, optional $\hat{\mathbf{x}}_t$, quality indicators q_t, and available actions $\mathcal{A}_t$.
Construct policy input:

$$z_t = \begin{cases} \emptyset, & \pi = \pi_0, \\ \tilde{\mathbf{x}}_t, & \pi = \pi_1, \\ [\tilde{\mathbf{x}}_t, \hat{\mathbf{x}}_t], & \pi = \pi_2. \end{cases}$$

for each $a \in \mathcal{A}_t$ **do**
Compute utility score $U_t(a \mid z_t; \mathcal{B}_\pi)$.
Estimate disparity $\widehat{\Delta}_{m,t}(a)$ and context-deficit penalty $C_t(a)$.
Compute

$$S_t(a) = U_t(a \mid z_t; \mathcal{B}_\pi) - \lambda \max(0, \widehat{\Delta}_{m,t}(a) - \epsilon) - \gamma C_t(a).$$

end for
Select $a_t \leftarrow \arg\max_{a \in \mathcal{A}_t} S_t(a)$.
Execute a_t and observe $r_t(a_t)$, outcome o_t, and group/flow label.
Update $\mathcal{B}_\pi$, $\mathcal{F}$, and $\mathcal{Q}$ using only selected-action feedback.
Append $\ell_t = \langle z_t, q_t, a_t, r_t(a_t), o_t, S_t(a_t) \rangle$ to $\mathcal{L}$.
end for
return $\mathcal{L}$, utility summaries, fairness summaries, and reporting datasets.

TABLE II
POLICY ALIASES USED IN ALGORITHMS 1 AND 2.

Alias	Context level	Implemented policy family
π_0	No context	`GNeuralUCB`, `EXPNeuralUCB`
π_1	Observed context	`CPursuitNeuralUCB`
π_2	Predictive context	`iCPursuitNeuralUCB`

D. Design Decisions

Bandits are used instead of full reinforcement learning because both testbeds provide bandit feedback: only the chosen diagnostic or routing action reveals an outcome. This study compares three policy families. Non-contextual MABs serve

TABLE III
APPLICATION MAPPING FOR THE SHARED METHOD.

Element	Clinical setting	Quantum setting
Context	site, cohort, service line, test signals, missingness	topology, load, fidelity, route state, capacity
Policy spectrum	no-context, observed-context, and predictive-context diagnostic allocation	no-context, contextual, and predictive-context routing policies
Action	test, retest, escalation, model-assisted triage	path, route, qubit allocation, retry policy
Reward	diagnostic utility, delay/cost penalty, error outcome	success, latency, fidelity, Oracle-normalized efficiency
Fairness gap	FNR/FPR, escalation delay, access imbalance	success-rate, latency, or service-class imbalance

TABLE IV
HIGH-LEVEL IMPLEMENTATION ARCHITECTURE.

Component	Responsibility
Environment abstraction	exposes domain state, actions, rewards, groups, context quality, and available policy inputs
Runner	executes no-context, contextual, and predictive-context policies over repeated rounds and seeds
Allocator/resource policy	controls scarce test, routing, qubit, or escalation capacity
EQUITAS mediator	applies fairness and context-quality adjustments to policy scores
Evaluator	aggregates run states into utility, regret, disparity, and context-spectrum evidence
State/reporting layer	persists model, runner, evaluator, fairness, provenance, and figure artifacts

as a baseline for action-only learning. Contextual bandits test whether observed context available before action selection improves utility and equity. Informative contextual bandits represent policies that recover, augment, or better weight missing context. The simulation-first clinical design avoids sensitive patient data while allowing controlled missingness and group-specific measurement noise. The quantum testbed reuses an existing, domain-grounded routing environment so that fairness analysis is built on top of realistic uncertainty rather than a toy network.

Fairness is measured as an online outcome property. In diagnostics, the target metrics are group-wise false-negative and false-positive gaps, along with utility and cost-sensitive reward. In quantum routing, service equity is measured through success-probability and latency gaps across flow groups. Mitigation mechanisms under consideration include fairness-regularized policy updates, group-aware calibration or thresholding, and missingness-aware context augmentation. These mechanisms will be evaluated as tradeoffs rather than as free improvements, because fairness constraints can reduce short-term reward while improving reliability and equity.

E. Planned Mitigation Strategy

The first mitigation family will treat fairness as a penalty or constraint in the policy-update loop. In the simplest form, the learner receives a utility reward and a disparity signal; policy updates are then adjusted when recent group-specific outcomes exceed an allowed gap. This makes fairness operational during learning instead of applying it only after the experiment. A second family will use missingness-aware context augmentation. If a group or flow class systematically receives incomplete context, the policy can either acquire additional features, impute missing indicators, or adjust confidence estimates so that uncertainty caused by measurement gaps is not mistaken for low expected value. A third family will test group-aware calibration or thresholding at the decision boundary. This is especially relevant for diagnostic-like escalation, where a small threshold shift can change false-negative rates.

The mitigation comparison will be conservative. The analysis will not claim that lower disparity is automatically better if the policy produces unsafe or unusable utility loss. Instead, each mitigation will be evaluated through utility–fairness tradeoff curves. The preferred policy is expected to be one that preserves most of the utility of contextual learning while reducing time-evolving disparity under missing context. This is why the study compares MAB, CMAB, and iCMAB policies under the same seeds, scenario settings, and logged artifacts: the point is to isolate the effect of context quality and fairness-aware updates rather than confounding algorithm choice with simulator configuration.

IV. IMPLEMENTATION AND REPRODUCIBILITY

A. Architecture

[modified: The implementation makes the shared runner in Algorithm 1 and the EQUITAS mediation procedure in Algorithm 2 executable without binding either one to a single domain. As shown in Fig. 3, each experiment is organized around six roles: an allocator, configuration, evaluator, runner, model, and visualizer. The environment supplies domain state, available actions, outcome semantics, and context-quality indicators. The allocator constrains scarce capacity before action selection, the runner executes one or more policy models, and the evaluator aggregates utility, regret, and disparity evidence. EQUITAS sits at the scoring boundary: it can mediate actions during an embedded run or audit saved quantum states without changing the original execution semantics.]

Table IV summarizes the component responsibilities. The architecture is domain-neutral but not domain-erasing: quantum experiments instantiate the environment with network topologies, link uncertainty, path decisions, allocator settings, and routing outcomes, while clinical experiments instantiate it with sites, cohorts, service lines, diagnostic actions, resource constraints, and diagnostic outcomes. Both environments emit state records with comparable context, action, outcome, group,

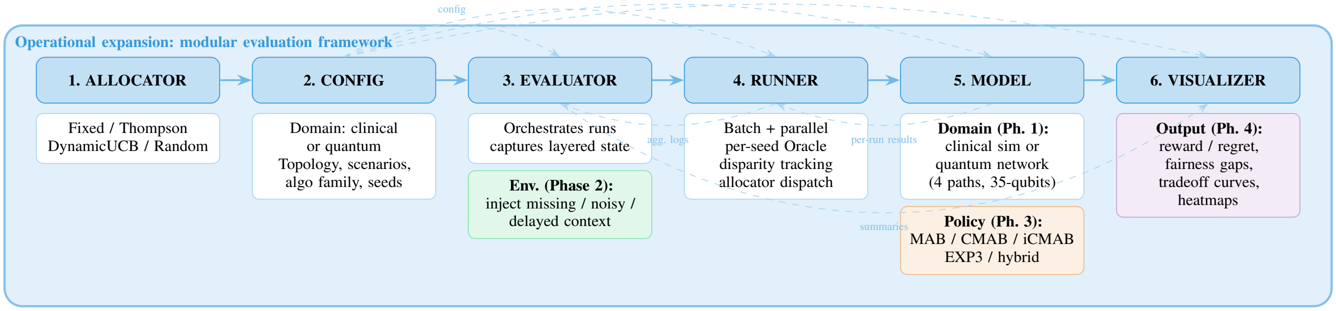

Fig. 3. Operational architecture for the shared method. The stack separates allocator, configuration, evaluator, runner, model, and visualizer responsibilities while preserving the intervention points from Fig. 2: context degradation in the evaluator environment, policy-family choice in the model layer, and utility–fairness tradeoff analysis in the visualizer. This separation is what allows the quantum and clinical findings to be reported through comparable utility and fairness artifacts.

allocator, and EQUITAS fields. That shared state contract supports the context-spectrum quantum results in Fig. 4 and the clinical EQUITAS supporting evidence in Appendix C.

B. State Persistence and Reporting

State is treated as a first-class research artifact rather than a temporary execution byproduct. Model state captures policy-level evidence, runner state captures round-level interactions for a configuration and seed, evaluator state captures cross-run summaries, and reporting state links derived datasets, figures, fairness summaries, and manifests to their source states. This layered record is essential for the preliminary findings: quantum saved states support post-run fairness analysis over existing experiments, while the embedded clinical runner produces state records that can be evaluated with the same fairness contract. As a result, claims about utility, context gain, and disparity can be traced back to the states that generated them.

C. Validation Strategy

Validation is organized around scientific contracts. A valid run must emit non-empty state records with context, action, outcome, group or flow label, allocator, and EQUITAS fields. A valid evaluator summary must preserve utility, oracle gap, success, reward, latency, and disparity metrics. A valid reporting package must link tables, figures, fairness summaries, and manifests to their source states. These checks do not replace scientific evaluation; they protect the evidence chain needed to interpret the preliminary quantum and clinical findings reported below.

V. Data Sets and Experimental Inputs

The experimental inputs are stateful simulator corpora rather than static classification tables. In the quantum domain, each run records a routing scenario with probabilistic link success, qubit-capacity constraints, allocator settings, time scale, model identity, reward, regret, Oracle-relative efficiency, failures, retries, and scenario winners. These fields support the context-spectrum findings in Fig. 4: the same corpus can compare no-context, contextual, and predictive-context policies while preserving the allocator and scenario conditions that produced each outcome.

Table V summarizes the quantum evidence base used in the preliminary results. The internal hybrid corpus provides the main within-framework comparison, while the standardized external corpus tests whether the same context-gain pattern appears across distinct paper testbeds. Because these corpora preserve evaluator and model state, the fairness layer can be applied without rerunning or redefining the legacy quantum pipeline.

Table VI gives the common state contract. The clinical testbed is simulation-first: each record represents a case context, site, cohort, service line, diagnostic action, allocation label, observed outcome, reward, success, latency, and context-quality signal. This design permits controlled evaluation of false-negative, false-positive, delay, and access gaps without requiring sensitive patient data during development. The bioinformatics EQUITAS package [2] provides separate supporting clinical-proxy evidence over RNA sequence type, length, and GC-content groups; its detailed quantitative role is reported in Appendix C.

The purpose of the shared schema is comparability, not semantic homogenization. Quantum features describe topology, link uncertainty, fidelity, capacity, and route state; clinical features describe site, cohort, service line, diagnostic signal, and resource allocation. Both domains, however, emit the same research primitives: context quality, chosen action, outcome, reward, group or flow label, allocator metadata, and provenance. This contract is what allows the preliminary results to tie utility gains and fairness gaps back to the same evaluator-state evidence.

VI. Ethics, Fairness, Privacy, and Security

The ethical risk follows directly from the decision model. A policy that maximizes aggregate reward can still be unacceptable if it shifts errors, latency, or resource denial onto groups with lower-quality context. In clinical routing, delayed testing or incomplete measurement can turn under-measurement into under-allocation. In quantum routing, unequal flow or service-class context can produce persistent differences in success probability, latency, or access to scarce path capacity.

TABLE V
PRIMARY QUANTUM DATASETS AND CORPORA USED BY THE PROJECT.

Corpus	Rows	Cols	Scope	Use in this project
Hybrid master dataset	4,920	30	internal hybrid framework	fairness-layer extension on top of allocator, scenario, reward, regret, and efficiency summaries
Standardized external papers corpus	8,060	32	Papers 2, 7, 8, and 12 at 4K/2K	cross-testbed comparison baseline for adding fairness and transfer analysis
Paper 2 standardized slice	2,450	30	15 nodes, 51 edges, 8 paths	stochastic-noise external benchmark
Paper 7 standardized slice	1,710	30	50 nodes, 141 edges, 15 paths	dense-network external benchmark
Paper 8 standardized slice	1,500	30	20 nodes, 19 edges, 8 paths	compact RL-routing external benchmark
Paper 12 standardized slice	2,400	30	100 nodes, 426 edges, 4 paths	fusion-network external benchmark

TABLE VI
SHARED STATE FIELDS USED FOR UTILITY AND FAIRNESS ANALYSIS.

Field type	Purpose
Group/flow label	subgroup or service class for disparity metrics
Context features	patient/test signals, link/load/fidelity estimates
Missingness flags	degraded, delayed, or unavailable context
Action choice	test, escalation, model, route/allocator decision
Outcome/reward	diagnostic & routing success, latency, or utility
Oracle/baseline	regret, gap & mitigation comparison

TABLE VII
ETHICAL DESIGN REQUIREMENTS AND IMPLEMENTATION RESPONSE.

Requirement	Implementation response
No aggregate-only reporting	utility, subgroup disparity metrics
Expose context quality	log missingness, noise, delay & measurement assumptions
Support accountability	preserve seeds, configs, states, manifests & run provenance
Protect sensitive data	simulation-first data until approved data are available
Run security relevance	quantum routing under unreliable or adversarial conditions

Fairness is therefore treated as an experimental design constraint rather than as a post-hoc descriptive statistic. The evaluation reports subgroup or service-class metrics, preserves context-quality indicators, and records the assumptions behind group definitions, missingness mechanisms, outcome gaps, and mitigation thresholds. This evidence is necessary for interpreting the preliminary results: higher average efficiency in Fig. 4 does not by itself establish equitable routing, and the clinical gaps in Appendix C require explicit utility–fairness tradeoff analysis.

Table VII summarizes how ethical concerns translate into design requirements. Privacy motivates the simulation-first clinical design until approved data are available. Security motivates the quantum scenarios because unreliable or adversarial conditions can alter who receives successful service. The framework does not claim to solve all privacy or security questions; it makes those assumptions explicit enough to vary, audit, and connect to outcome evidence.

The EQUITAS preliminary evidence reinforces the same ethical point: under-represented sequence contexts, such as low-GC examples, required explicit augmentation and re-auditing before fairness claims were defensible. The ethical standard for the final analysis is therefore stronger than "the algorithm works." The final system should make visible when a policy is accurate but inequitable, robust but unfair to a service class, or fairer only because it sacrifices too much utility to be operationally useful. This tradeoff framing is important for applied data science: fairness is part of the evidence required to justify deployment or to reject a policy as unsafe.

VII. Preliminary Results

A. Quantum Preliminary Results

[modified: The preliminary results are evidence for the proposed method rather than deployment-level fairness claims. The quantum results address RQ1 by isolating the contribution of context in a preserved legacy routing corpus. Because these runs predate online EQUITAS mediation in the quantum loop, they define the utility frontier that a fairness-aware policy should preserve while reducing service disparities. Appendix D summarizes the source evidence; the main text reports the context-spectrum results needed to connect the algorithmic transition to observed performance.]

Figure 4 separates absolute routing quality from the incremental value of added context. The uplift panels show that predictive context is consistently the strongest improvement family, while the size of the gain varies materially by threat regime rather than remaining constant across scenarios.

B. Clinical Preliminary Results

[modified: The clinical results provide two complementary signals. First, the embedded clinical evaluator shows that an EQUITAS-aware allocator can improve oracle-normalized efficiency from 75.09% to 91.23%, raise mean success from 50.0% to 72.5%, and reduce the latency parity gap from 18.0 to 10.17 relative to the baseline on the synthetic fixture. Second, the EQUITAS bioinformatics source package supplies a diagnostic-relevant proxy for context restoration: lower context corresponds to structure prediction without thermodynamic/context-aware scoring, while richer context corresponds to Nussinov/MFE feature augmentation, Synthetic LowGC augmentation, and subgroup-aware auditing.

TABLE VIII
CONTEXT-SPECTRUM SUMMARY FROM THE EXISTING GA CORPORA. GAINS ARE MEASURED AGAINST THE NO-CONTEXT FAMILY WITHIN EACH CORPUS.

Context level	Representative model(s)	Internal mean	Δ vs. no ctx.	External mean	Δ vs. no ctx.
No context	`GNeuralUCB`, `EXPNeuralUCB`	82.39%	—	62.83%	—
Context	`CPursuitNeuralUCB`	84.70%	+2.31 pp	64.11%	+1.28 pp
Predictive context	`iCPursuitNeuralUCB`	**86.13%**	**+3.74 pp**	**65.66%**	**+2.83 pp**

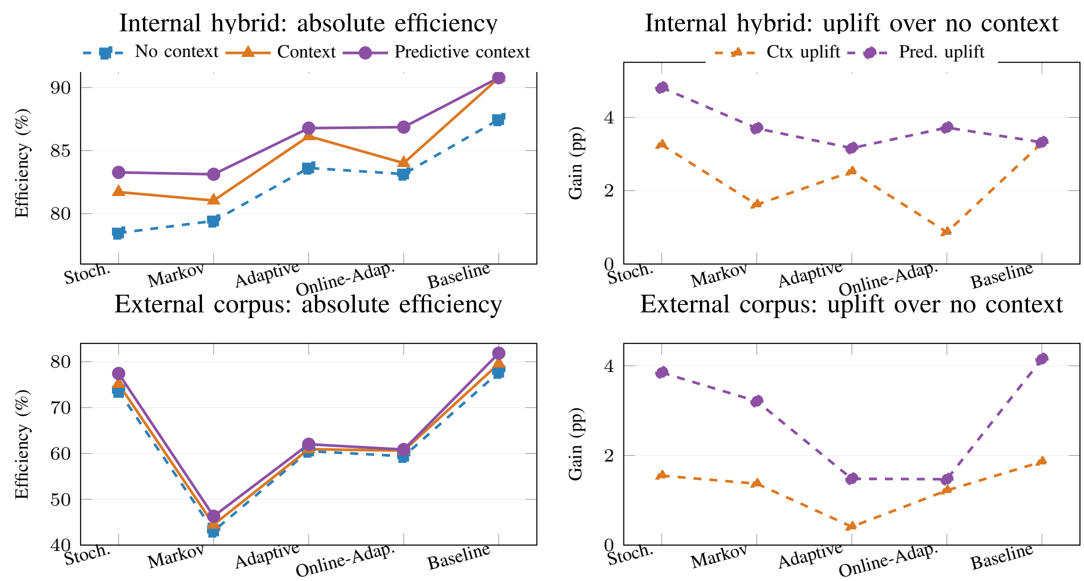

Fig. 4. Quantum context-spectrum results decomposed into absolute performance and marginal uplift. The left column shows the absolute efficiency trajectories for no-context, contextual, and predictive-context families in the internal and external corpora. The right column shows how much contextual and predictive context improve over the corresponding no-context baseline in each scenario. This makes the scientific point explicit: richer context helps everywhere, but the size of the gain is scenario dependent and is usually largest for the predictive-context family.

TABLE IX
CROSS-TESTBED PREDICTIVE-CONTEXT UPLIFT IN THE STANDARDIZED EXTERNAL CORPUS.

Testbed	No ctx.	Context	Pred. ctx.	Pred. gain
Paper 2	72.11	73.18	74.34	+2.23
Paper 7	72.87	73.44	78.39	+5.52
Paper 8	70.05	73.69	73.95	+3.90
Paper 12	41.69	42.23	42.54	+0.85

TABLE X
AGGREGATE CONTEXT-GAIN COMPARISON ACROSS THE TWO PRELIMINARY EVIDENCE BASES.

Testbed	Lower ctx.	Richer ctx.	Gain
Quantum routing	82.39%	86.13%	+3.74 pp
Clinical bioinformatics	14.96 pairs	66.07 pairs	+51.11 pairs

Appendix C reports the detailed source evidence; the main results focus on the quantitative pattern and its connection to the routing method.]

Figure 5 shows that quantum gains are steady and incremental across context levels, whereas the clinical proxy is dominated by a large first-stage recovery when domain-relevant context is restored and then a smaller second-stage gain associated with tuning and fairness-aware augmentation.

Appendix D gives the source-level quantum evidence behind these results. The validated corpus already contains model–scenario–allocator–scale evaluations across the internal hybrid framework and the 8,060-row standardized external corpus. In the internal hybrid corpus, `iCPursuitNeuralUCB` defines the current utility frontier at 86.13% mean efficiency, followed by `CPursuitNeuralUCB` at 84.70%. These results identify which policy families should be extended with EQUITAS mediation first.

The context-spectrum view in Table VIII and Figure 4 directly addresses RQ1. The no-context condition captures reward-only or non-context baselines, the context condition captures methods that use observed contextual structure, and the predictive-context condition captures informative or forecasting-aware extensions. In both internal and external corpora, performance rises as the method family moves from no context to context to predictive context. The external standardized corpus also shows that gains vary by testbed structure: `iCPursuitNeuralUCB` reaches approximately 74.34% on Paper 2, 78.39% on Paper 7, 73.95% on Paper 8, and 42.54% on Paper 12. This variation supports the need for

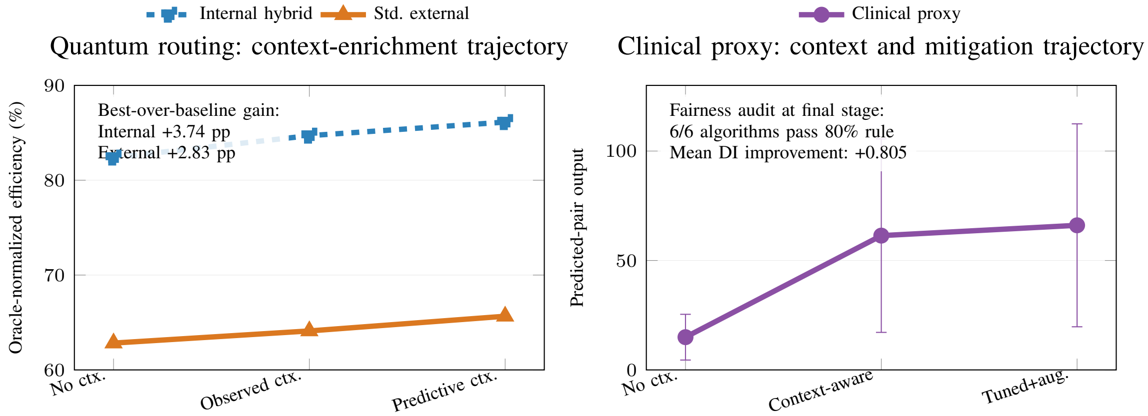

Fig. 5. Cross-testbed context-enrichment trajectories. The left panel shows that quantum-routing performance improves monotonically as the policy moves from no context to observed context to predictive context, with consistent gains in both the internal hybrid corpus and the standardized external corpus. The right panel separates two clinical effects that a single normalized index would obscure: restoring domain-relevant context produces the large primary gain (14.96 to 61.36 predicted pairs), while tuning and fairness-oriented augmentation provide a smaller secondary improvement (66.07 at the final stage) together with improved parity indicators. This decomposition distinguishes pure context recovery from later mitigation or refinement.

cross-testbed fairness analysis rather than a single topology-specific claim.

The clinical evidence addresses the same context hypothesis from a biomedical angle. The EQUITAS package shows that restoring domain-relevant context changes prediction behavior substantially, while under-represented-context augmentation improves parity indicators. These results are preliminary because they come from a clinical proxy rather than the final sequential diagnostic simulator. They are nevertheless methodologically useful: they show that the EQUITAS framing can represent context quality, group representation, utility, and fairness evidence in a form that can later be routed through the same bandit-style evaluator used for quantum experiments.

[modified: Taken together, the preliminary findings support three cautious claims. First, richer context improves utility in both the quantum corpus and the clinical proxy, but the size of the gain is domain- and scenario-dependent. Second, the embedded clinical fixture suggests that EQUITAS-style mediation can improve utility and parity indicators simultaneously under controlled conditions. Third, saved states, fairness sidecars, and reporting manifests make the evidence traceable enough to support the planned time-evolving disparity analysis for RQ2 and RQ3. These findings do not yet identify a universally fairest policy; they establish the empirical and software basis for testing whether Eq. 5 preserves utility while reducing disparity.]

VIII. Discussion and Limitations

[newtext: The results suggest that context quality is not only a utility variable but also a fairness mechanism. In both domains, the learner makes sequential choices from incomplete evidence, and the groups or flows with less reliable context are the ones most exposed to repeated allocation error. The shared runner, allocator, state, and reporting contracts are therefore central to the contribution: they allow clinical and quantum experiments to differ in domain semantics while still producing comparable utility, disparity, and provenance evidence. This design makes fairness mediation an executable research object rather than a post hoc table added after performance analysis.]

[newtext: Several limitations qualify the current evidence. The quantum results come from simulator and saved-state corpora, and the current fairness layer audits those states before full online quantum mediation has been evaluated. The clinical results use a synthetic fixture and a bioinformatics proxy rather than approved patient-level data, so they support mechanism testing rather than clinical deployment claims. Group and service-class definitions are also experimental choices, not neutral facts; different definitions could change the observed disparity curves. Finally, the current analysis emphasizes effect direction and traceability, while the final study should add larger sequential runs, uncertainty estimates, ablations, and sensitivity analysis over mitigation weights.]

IX. Future Work

[modified: Future work will complete the experimental transfer test implied by RQ4. The first step is to run EQUITAS mediation online in the quantum evaluator so that success-probability, latency, and service-class gaps are measured during learning rather than only after legacy runs. The second step is to extend the clinical diagnostic simulator with controlled missingness, group-dependent measurement noise, diagnostic action choices, and false-negative or false-positive outcomes. The third step is to compare utility-only, missingness-aware, and fairness-regularized policies under shared seeds and report utility–fairness tradeoff curves rather than single aggregate scores.]

X. Conclusion

[modified: This work presented a fairness-aware bandit framework for routing scarce resources under missing, delayed, noisy, and group-dependent context. The central contribution is a reusable decision spectrum that connects non-contextual, contextual, and predictive-context bandits to a shared EQUITAS-

mediated fairness layer. The quantum evidence shows consistent utility gains from context enrichment across internal and external routing corpora, while the clinical evidence shows that EQUITAS-aware allocation and context restoration can improve controlled utility and parity indicators.]

[modified: The broader implication is that fairness in sequential routing cannot be assessed from aggregate reward alone. It requires stateful evidence about what context was available, which action was selected, which group or flow was affected, and how utility and disparity evolved over time. By preserving these states across clinical and quantum settings, the framework establishes a reproducible basis for testing whether fairness mediation can reduce allocation harm without discarding the utility benefits of contextual learning.]

REFERENCES

[1] P. Z. Garcia, "Threat-aware evaluation of quantum entanglement routing and qubit allocation via hybrid contextual bandits," 2026, unpublished manuscript (GA-Work internal draft), Rochester Institute of Technology.

[2] ——, "Equitable bioinformatics: Enhancing diagnostic decision-making through RNA and biomarker data," 2025, course paper (ISTE 780 Phase 4), Rochester Institute of Technology.

[3] P. Auer, N. Cesa-Bianchi, and P. Fischer, "Finite-time analysis of the multiarmed bandit problem," *Machine Learning*, vol. 47, no. 2–3, pp. 235–256, 2002.

[4] D. Russo, B. Van Roy, A. Kazerouni, I. Osband, and Z. Wen, "A tutorial on Thompson sampling," *Foundations and Trends® in Machine Learning*, vol. 11, no. 1, pp. 1–96, 2018.

[5] T. Lattimore and C. Szepesvári, *Bandit Algorithms*. Cambridge University Press, 2020.

[6] L. Li, W. Chu, J. Langford, and R. E. Schapire, "A contextual-bandit approach to personalized news article recommendation," in *Proceedings of the 19th International Conference on World Wide Web*, 2010, pp. 661–670.

[7] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári, "Improved algorithms for linear stochastic bandits," in *Advances in Neural Information Processing Systems*, 2011.

[8] A. Agarwal, D. Hsu, S. Kale, J. Langford, L. Li, and R. E. Schapire, "Taming the monster: A fast and simple algorithm for contextual bandits," in *Proceedings of the 31st International Conference on Machine Learning*, 2014.

[9] M. Dudík, J. Langford, and L. Li, "Doubly robust policy evaluation and learning," in *Proceedings of the 28th International Conference on Machine Learning*, 2011.

[10] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Advances in Neural Information Processing Systems*, 2016.

[11] M. Joseph, M. Kearns, J. Morgenstern, and A. Roth, "Fairness in learning: Classic and contextual bandits," in *Advances in Neural Information Processing Systems*, 2016.

[12] X. Chen, A. Zheng, Z. Zhou, and N. B. Shah, "Fair contextual multi-armed bandits: Theory and experiments," in *UAI*, 2020.

[13] B. Metevier, S. Giguere, S. Brockman, A. Kobren, Y. Brun, E. Brunskill, and P. Thomas, "Offline contextual bandits with high probability fairness guarantees," in *NeurIPS*, 2019.

[14] A. Garivier and E. Moulines, "On UCB policies for non-stationary bandit problems," *arXiv preprint arXiv:0805.3415*, 2008.

[15] O. Besbes, Y. Gur, and A. Zeevi, "Stochastic multi-armed-bandit problem with non-stationary rewards," in *NeurIPS*, 2014.

[16] D. Bouneffouf, I. Rish, G. Cecchi, and R. Féraud, "Context attentive bandits: Contextual bandit with restricted context," in *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence*, 2017.

[17] D. Park and M. S. Faradonbeh, "Efficient algorithms for learning to control bandits with unobserved contexts," *arXiv preprint arXiv:2210.15668*, 2022.

[18] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science*, vol. 366, no. 6464, pp. 447–453, 2019.

[19] S. Wehner, D. Elkouss, and R. Hanson, "Quantum internet: A vision for the road ahead," *Science*, vol. 362, no. 6412, p. eaam9288, 2018.

[20] M. Pant, H. Krovi, D. Englund, L. Gyongyosi, M. Babroli *et al.*, "Routing entanglement in the quantum internet," *npj Quantum Information*, vol. 5, no. 25, 2019.

[21] M. Caleffi, "Optimal routing for quantum networks," *IEEE Access*, vol. 5, pp. 22 299–22 312, 2017.

Appendix A
Quantum Routing Motivating Example

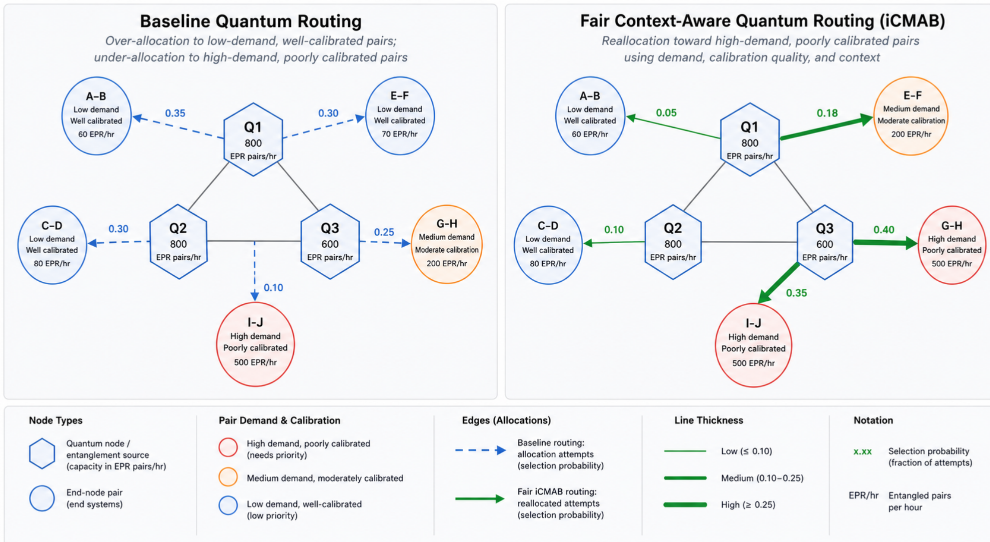

Fig. 6. Quantum routing example corresponding to the clinical routing pattern shown in Figure 1. The left side shows baseline routing and the right side shows context-aware fairness allocation illustrating how the same partial-feedback and context-quality issue appears in quantum network routing.

Appendix B
Operational Evidence-Flow Architecture

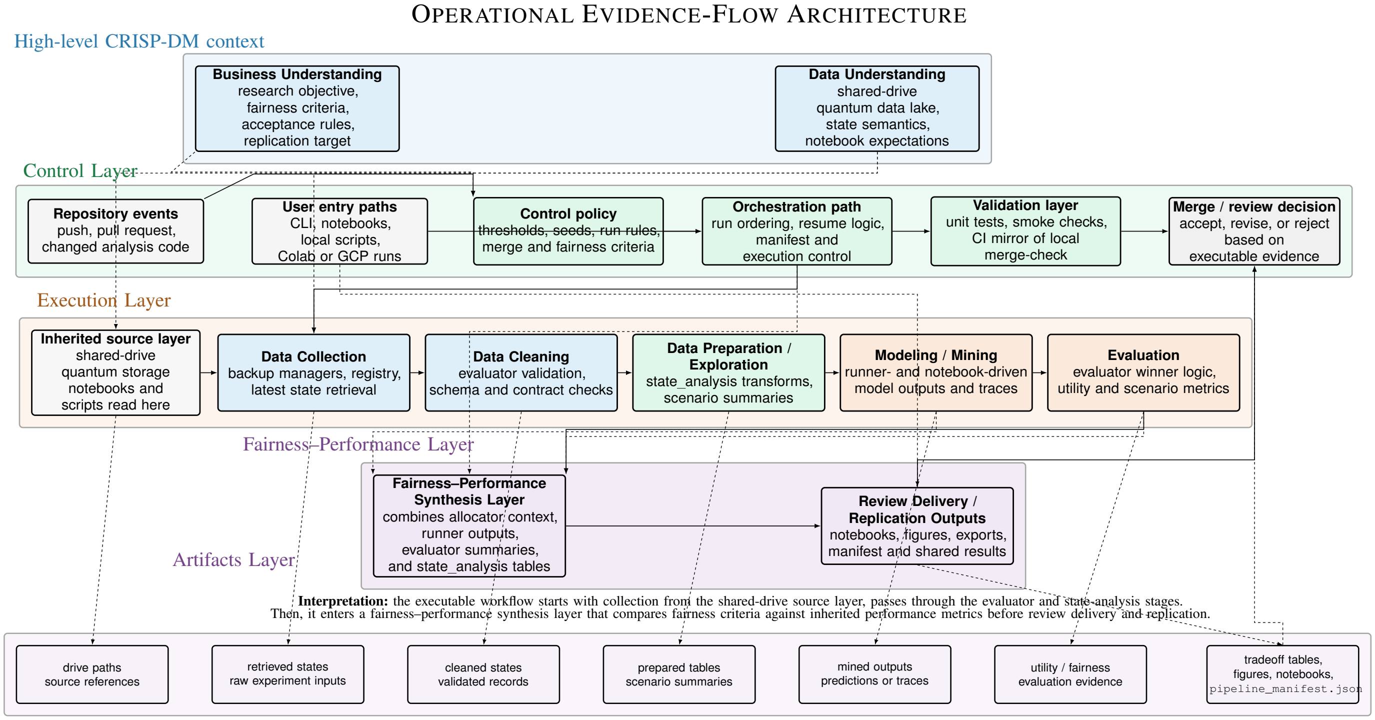

Fig. 7. Full evidence-flow architecture. Domain context enters through an environment abstraction, runner/model/allocator decisions produce stateful outcomes, the EQUITAS layer converts those outcomes into utility and fairness evidence, and the reporting layer emits datasets, figures, manifests, and reproducibility artifacts. The same flow supports clinical simulation and quantum routing without forcing either domain to imitate the other.

Appendix C
EQUITAS Bioinformatics Evidence Package

The EQUITAS bioinformatics source package is retained as supporting clinical proxy evidence rather than as the primary embedded clinical experiment. It shows how restoring domain context and repairing under-represented sequence contexts can change both prediction behavior and fairness indicators.

TABLE XI
Preliminary clinical results from the EQUITAS bioinformatics evidence package.

Evidence source	Current quantitative signal	Why it matters	Role in this study
EQUITAS corpus	dataset expanded from 10 to 20 RNA sequences, including 10 Synthetic LowGC sequences	makes clinical-like context gaps measurable in a controlled biomedical proxy task	informs missingness and group-representation design for clinical routing
Enhanced Nussinov tuning	Optuna search over 30 trials produced an optimized objective of 56.44 and mean predicted-pair output of 66.07 $\pm$ 46.36	shows that richer structural context can change diagnostic-relevant prediction behavior	motivates lower-context versus richer-context policy comparisons
Context ablation	thermodynamic/context-aware scoring reported 61.36 $\pm$ 44.21 predicted pairs versus 14.96 $\pm$ 10.45 without that context	supports the context-gain hypothesis in a biomedical proxy task	anchors the clinical context-restoration evidence used in the main text
Fairness audit	post-augmentation mitigation achieved 6/6 algorithm compliance with the 80% rule, DI ratios of 1.000, and +0.805 average DI improvement	shows that representation repair can improve fairness indicators, not only utility	supplies supporting fairness evidence before full sequential bandit experiments

Appendix D
Quantum Preliminary Evidence Package

The inherited quantum-routing evidence package is retained as supporting source evidence for the main context-spectrum results. It documents why the quantum corpus is strong enough to support preliminary context-gain claims before full EQUITAS-mediated service-equity experiments are completed.

TABLE XII
Preliminary quantum results available before fairness-layer extension.

Evidence source	Current quantitative signal	Why it matters	Role in this study
Threat-aware quantum manuscript	pursuit–neural hybrids outperform non-contextual baselines by roughly 18–24 percentage points in scenario-aggregated efficiency	shows that context-rich routing materially improves utility under realistic threats	establishes the performance baseline to which the fairness layer will be added
Threat-aware quantum manuscript	worst-case performance floors remain above 85% under stochastic threats for the strongest hybrid family	suggests the best policies remain stable in benign-but-noisy settings	provides a natural starting point for testing whether fairness can be improved without breaking robust utility
Threat-aware quantum manuscript	replay-capacity increases can trigger 22–30 percentage-point efficiency collapses under adaptive adversaries	shows that resource policy, not only algorithm choice, shapes deployment quality	motivates fairness-aware evaluation of allocator and capacity decisions, not just bandit model choice
Standardized external corpus	8,060 rows across 4 testbeds, 4 allocators, 5 threat scenarios, 3 scales, and 5 routing models including Oracle	demonstrates breadth of already available quantum evidence	supports cross-testbed fairness benchmarking once subgroup/service labels are defined
Internal hybrid corpus	`iCPursuitNeuralUCB` currently has the highest mean efficiency in the internal hybrid dataset (86.13%)	identifies the strongest current utility baseline in the internal framework	provides the leading candidate for fairness-aware extension and ablation

Appendix E
Algorithmic Context-Spectrum Trace

[newtext: This appendix keeps the wider side-by-side trace for space reasons. It expands Algorithm 1 by showing where the three policy families diverge in code-level evidence and interpretation.]

TABLE XIII
Algorithmic transition across the context spectrum.

MAB: no context	CMAB: observed context	iCMAB: predictive context
Input: $z_t = \emptyset$ 1) Track action pulls and rewards. 2) Score each action from reward history only. 3) Select the action with highest uncertainty-adjusted reward. 4) Update only the selected action. **Code family:** `GNeuralUCB`, `EXPNeuralUCB` **Role:** baseline for measuring the cost of missing context.	**Input:** $z_t = \tilde{\mathbf{x}}_t$ 1) Observe available context. 2) Condition each action score on $\tilde{\mathbf{x}}_t$. 3) Select the action with highest context-conditioned utility. 4) Update the selected action using observed context and reward. **Code family:** `CPursuitNeuralUCB` **Role:** tests whether observed context improves utility and disparity.	**Input:** $z_t = [\tilde{\mathbf{x}}_t, \hat{\mathbf{x}}_t]$ 1) Observe context and estimate missing or future context. 2) Augment the scorer input with $\hat{\mathbf{x}}_t$. 3) Select the action with highest augmented-context utility. 4) Update the selected action using augmented evidence and reward. **Code family:** `iCPursuitNeuralUCB` **Role:** tests whether context recovery further reduces disparity risk.